

Latent Heterogeneity in Health Care Utilization Counts Using Negative Binomial Mixtures

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Abstract

We assess whether finite Negative Binomial mixture models recover latent utilization groups more accurately than clustering in overdispersed count data. We generated a synthetic dataset of 5,000 beneficiaries with five health-care utilization variables, structured into four latent groups and calibrated to Brazilian regulatory benchmarks. Model parameters were estimated by maximum likelihood using Expectation-Maximization algorithm. The results suggest that finite Negative Binomial mixture models provide a more suitable approach to identifying latent utilization profiles in overdispersed health-care count data.

Keywords: finite mixture models, negative binomial distribution, health-care utilization.

Health-care utilization data are typically characterized by discreteness, asymmetry and overdispersion, often exhibiting substantial variability between individuals with low and high levels of service use. These features challenge the adequacy of purely geometric clustering approaches, such as K-means, particularly when the underlying population comprises latent subgroups. Finite mixture models provide a probabilistic framework for representing unobserved heterogeneity by assuming that the observed population arises from a combination of latent components [1, 2].

In this work, we investigate whether a finite mixture model based on Negative Binomial components can recover latent utilization profiles more accurately than K-means. We generated synthetic [3] to emulate health-care utilization in Brazil, following the simulation approach of Deb and Trivedi [4] for modelling health-care demand via finite mixtures. The data consists of 5,000 beneficiaries, structured into four latent groups and described by five count variables: *medical consultations*, *emergency visits*, *diagnostic exams*, *hospitalizations* and *therapies*. The marginal distributions of the simulated variables were calibrated using aggregate indicators from the Brazilian National Regulatory Agency for Private Health Insurance and Plans. Since the true latent group membership is known by construction, the performance of each clustering method can be assessed using clustering performance was evaluated using the Adjusted Rand Index (ARI) [8] and Classification accuracy [9].

The proposed model assumes that each latent component follows a product of independent Negative Binomial distributions [5], one for each utilization variable. Hospitalization and therapy variables exhibit structural excess of zeros [6], accommodated within each component. Parameter estimation was conducted using maximum likelihood via the Expectation-Maximization algorithm [7],

with the number of components selected according to the Bayesian Information Criterion (BIC). For comparison, K-means clustering was applied to log-transformed and standardized count data, with the number of clusters selected using the Silhouette criterion. These preprocessing steps are necessary because K-means relies on Euclidean distances, which are sensitive to scale differences and distributional skewness [10].

The best performance was obtained by the Negative Binomial mixture model when all five variables were included. In this specification, the BIC selected four components, matching the true number of latent groups. The model attained $\text{ARI} = 0.729$, $\text{NMI} = 0.594$, $\text{accuracy} = 0.877$ and $\text{purity} = 0.861$, outperforming K-means, which achieved $\text{ARI} = 0.673$ and $\text{accuracy} = 0.815$. The inclusion of rare but informative variables, particularly therapies, improved the recovery of the latent structure.

These results provide evidence that, in overdispersed count data with latent heterogeneity, Negative Binomial mixture models constitute a more suitable framework than traditional clustering methods for recovering hidden utilization profiles. This is a methodological study and does not aim to classify real beneficiaries or detect operational anomalies.

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